Compound Symmetry versus Separable AR(1): Why We Changed the Visit-to-Visit Covariance Model

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Purpose. This note explains one specific change in how our simulations model the correlation structure of a patient's repeated symptom-component scores: replacing a **compound-symmetry** (CS) correlation matrix with a **separable AR(1)** one, and, cross-factor, replacing a constant cross-component correlation with one that decays with the separable AR(1) factor. It is written for a reader with statistical training (correlation matrices, variance components, mixed models) but no prior exposure to this project's notation, in the style of our more code-focused `docs/53`/`docs/55` notes. Unlike `docs/35-separable-response-covariance.md`, the full technical derivation this note draws on, this is meant to be read once, stand alone, and go as far as the Kronecker/separability argument and its consequence for the feasible biomarker-effect range, without reproducing every derivation.

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Notation

Symbol	Meaning
w_t	calendar week of visit t (Hybrid design: 4, 8, 9, 10, 11, 12, 16, 20)
ρ	within-component AR(1) autocorrelation per week (0.7 here)
ρ_{cs}	compound-symmetry correlation, constant across all visit pairs (0.8 as published by Hendrickson et al.; 0.7 in the package default)
c_1	cross-component correlation at zero lag (same visit), under the separable form (0.2)
$c_\times$	the published process's constant cross-component correlation at every lag, same or different visit (0.1)
TV, PB, BR	the three latent response components (natural course, placebo belief, biological drug response)
K_p	the $p \times p$ cross-component correlation matrix at a single, shared timepoint ($p = 3$ here)
K_t	the $n \times n$ within-component correlation matrix across the n visits
$\otimes$	Kronecker product
Σ	the full $(np) \times (np)$ correlation matrix over every (component, visit) pair

1. Summary

The question. How correlated is a patient's score on one component (say, the drug-response component, BR) at two different visits, as a function of how many weeks apart those visits are? And separately: how correlated are *different* components (say TV and PB) with each other, at the same visit and at different visits?

Compound symmetry (the published process). Within a component, every pair of visits shares one correlation, ρ_{cs}, regardless of lag. Across components, every pair of visits also shares one constant, $c_\times$, again regardless of lag, except at the same visit, where it is a different constant c_1. Formally, the within-component block is

$$K_t^{CS} = (1 - \rho_{cs})\, I_n + \rho_{cs}\, J_n,$$

where I_n is the identity and J_n the all-ones matrix: every off-diagonal entry equals ρ_{cs}, independent of which pair of visits it connects. This is the classical **equicorrelation** or **exchangeable** structure: visit order and spacing carry no information.

Separable AR(1) (the refined process). Within a component, correlation decays geometrically with the gap in weeks:

$$[K_t]_{t,s} = \rho^{|w_t - w_s|}.$$

Across components, the correlation at any pair of visits, same or different, is the cross-component correlation K_p (constant c_1 off-diagonal) **multiplied by the same time-decay factor** that governs the within-component case:

$$\Sigma_{(c,t),(c',s)} = [K_p]_{c,c'} \times \rho^{|w_t - w_s|}.$$

This multiplicative structure — one factor for "which pair of components" and a separate, shared factor for "how far apart in time" — is what **separable** means here, made precise in Section 3.

Why it matters. The compound-symmetry structure imposes a hard ceiling on how strong the biomarker-response coupling can be before the joint distribution becomes mathematically infeasible (Section 4). Under Hendrickson's own published value, that ceiling is 0.256 (on a correlation scale where 1 is deterministic). Under separable AR(1) at $\rho = 0.7$, it is approximately 0.474-0.480 — nearly double. The clinical motivation (symptom scores recorded a week apart should resemble each other more than scores sixteen weeks apart) and the statistical motivation (a higher, more defensible ceiling; exact coherence with the fitted analysis model's residual structure) point the same direction, which is why both were adopted together.

2. Compound symmetry and AR(1) as covariance models

Restricting attention to one component for now (say BR, at the $n = 8$ Hybrid-design visits), a correlation matrix is a statement about how *any* two visits for the same patient relate. Two classical structures dominate the repeated-measures literature:

Compound symmetry / exchangeable. $\mathrm{Cor}(Y_t, Y_s) = \rho_{cs}$ for every $t \neq s$. This is exactly the correlation structure induced by a **random-intercept-only** model: $Y_{it} = b_i + \varepsilon_{it}$, with $b_i \sim N(0, \sigma_b^2)$ a patient-level random effect and $\varepsilon_{it} \sim N(0, \sigma_e^2)$ i.i.d. Then $\rho_{cs} = \sigma_b^2/(\sigma_b^2 + \sigma_e^2)$, the intraclass correlation (ICC). Every pair of visits is equally correlated because the *only* source of within-patient dependence is a single, time-invariant patient effect; nothing in the model lets two visits that happen to be close in time resemble each other any more than two visits far apart.

AR(1). $\mathrm{Cor}(Y_t, Y_s) = \rho^{|w_t - w_s|}$. This is the correlation induced by a first-order autoregressive process in continuous time: $Y_{it} = \mu_i + u_{it}$, where u_{it} follows $u_{i,t} = \rho^{\Delta} u_{i,t-1} + \text{noise}$ for a time gap Δ. Two visits close together inherit most of each other's deviation from the patient's mean trend; two visits far apart inherit almost none of it. This is the structure `nlme::corCAR1()` fits (the continuous-time generalization of discrete-lag AR(1), needed here because Hybrid-design visits are unevenly spaced).

Neither is more "correct" in the abstract; they encode different substantive claims about *why* repeated measurements on the same patient are correlated. Compound symmetry says: there is a stable patient trait that shifts every visit equally. AR(1) says: deviations from trend are serially correlated and fade with elapsed time. For a fluctuating symptom measure recorded over 4 to 20 weeks, the second claim is the more clinically defensible one — and, not coincidentally, it is also the residual structure the project's recommended analysis model (`corCAR1`, random intercept) actually assumes when fitting real or simulated data. Using CS to *generate* the data while fitting an AR(1) residual structure to *analyze* it is an avoidable data-generating-process/analysis-model mismatch; Section 5 returns to this coherence point.

3. What "separable" means, precisely

The full problem is not one component at n visits, but **three** latent components (TV, PB, BR) each observed at the same n visits, plus their correlation with a biomarker measured once. Restricting to the three components, the full correlation structure is an $(np) \times (np)$ matrix Σ indexed by pairs (component, visit). A **separable** (equivalently, Kronecker-product) covariance structure is one that factors as

$$\Sigma = K_p \otimes K_t,$$

where K_p is $p \times p$ (here $p = 3$, the component-to-component correlation at a single shared timepoint) and K_t is $n \times n$ (the within-component, cross-visit correlation). Written entrywise, for components c, c' and visits t, s:

$$\Sigma_{(c,t),(c',s)} = [K_p]_{c,c'} \cdot [K_t]_{t,s}.$$

The substantive claim this encodes is that **the correlation between any two (component, visit) pairs splits into two independent factors**: one that depends only on *which pair of components* is involved, and one that depends only on *which pair of visits* is involved, with no interaction between the two. Equivalently: the *ratio* of cross-component to within-component correlation is the same at every lag. Under the refined process, K_t is the AR(1) matrix of Section 2 ($[K_t]_{t,s} = \rho^{|w_t - w_s|}$), and K_p has 1's on the diagonal and c_1 off it, so

$$\Sigma_{(c,t),(c',s)} = \begin{cases} \rho^{|w_t - w_s|} & c = c' \\ c_1\, \rho^{|w_t - w_s|} & c \neq c'. \end{cases}$$

Compare this with the published process. There, the within-component block uses K_t^{CS} (Section 2), and the cross-component block uses a *constant* $c_\times$ at every lag except zero, where it is c_1 instead. That is **not** separable in the sense defined above: the cross-component correlation does not scale with the *same* decay function that governs the within-component block (indeed it has no decay at all once $t \neq s$), so no single $K_p \otimes K_t$ factorization reproduces it. `docs/35-separable-response-covariance.md` (Section 4) works out the published structure explicitly as a sum of two terms, $K_p^{\text{person}} \otimes J_n + K_p^{\text{occasion}} \otimes (I_n - \frac{1}{n} J_n)$ in one parameterization — a valid correlation structure, but a strictly more complicated, non-separable one, with person-level and occasion-level cross-component coupling ratios that differ (0.143 and 0.333 at $\rho_{cs} = 0.7$; *quoted*, `docs/35`).

Why adopt the simpler, separable form. Four reasons, elaborated in Section 5: it is valid (positive definite) for *every* visit schedule, not only ones checked case by case; it needs one cross-component parameter instead of two; it is the standard model for multivariate repeated measures, with an established literature on estimation and on testing separability itself; and, with constant component standard deviations, it makes the *summed* observed score (the sum of the three latent components, which is what gets analyzed) exactly a random-intercept-plus-AR(1) process — the same structure the recommended analysis model fits.

4. A worked numeric comparison

The Hybrid design's eight visits fall at weeks 4, 8, 9, 10, 11, 12, 16 and 20. The table compares, within one component, what each correlation model predicts for a patient's own score at representative visit pairs.

Table 1: Within-component correlation under each structure (*derived*)

Visit pair	Gap (weeks)	CS: ρ_{cs}	AR(1): ρ^{gap}, $\rho = 0.7$
Week 11 vs. week 12	1	0.80	0.700
Week 4 vs. week 8	4	0.80	0.240
Week 4 vs. week 12	8	0.80	0.058
Week 4 vs. week 20	16	0.80	0.0033

Every CS entry is identical by construction; AR(1) shrinks by a factor of ρ for each additional week of gap. At the largest gap in the design, CS overstates the correlation by a factor of roughly 240 relative to AR(1).

For the **cross-component** comparison, take BR and PB at the same visit (lag 0) and at an 8-week gap. Separable AR(1) gives $c_1\rho^0 = 0.2$ at lag 0 and $c_1\rho^8 = 0.2 \times 0.7^8 = 0.2 \times 0.0576 = 0.0115$ at an 8-week gap. The published process gives $c_1 = 0.2$ at lag 0 (same as separable AR(1), since both use c_1 at zero lag) but a flat $c_\times = 0.1$ at *every* positive lag, including the 8-week one — 0.1 against separable AR(1)'s 0.0115, an almost-ninefold overstatement at that gap, and with no further decay at longer gaps either.

5. Why we want separability, in more depth

Validity for every schedule, in closed form. A compound-symmetric or separable correlation matrix's eigenvalues have closed forms that do not depend on how many visits there are or how they are spaced

(`docs/35`, Section 4.2-4.3): for the separable form, the matrix is positive definite whenever K_p and K_t individually are, which holds for any $\rho \in (-1, 1)$ and any c_1 inside an explicit, n-independent range. The published process's non-separable structure instead requires checking positive-definiteness numerically, design by design, and (as `docs/55`/`docs/61` document) the shipped code's response to a violation was to silently repair the matrix rather than reject the input.

An exact, checkable ceiling. Section 6 of `docs/35` derives the largest biomarker-response coupling c^*_{bm} the joint distribution can support, as a closed-form function of ρ (or ρ_{cs}) and the visit schedule — not a number that must be rediscovered by trial and error for each new design. Numerically: 0.256 under the published compound-symmetry structure; 0.440 at $\rho_{cs} = 0.8$ under separable AR(1); 0.474 (0.480 in the Hybrid design specifically) at $\rho = 0.7$ (*quoted*, `docs/series/ch10-refined-process.md`).

Fewer parameters, standard structure. One cross-component constant (c_1) instead of two ($c_1, c_\times$); Kronecker-product covariances are an established model for multivariate repeated measures, with known estimation theory and known tests of the separability assumption itself (`docs/35`, Section 5, items 4 and 9).

Coherence with the analysis model. With the three components' standard deviations held constant across visits, the *sum* of the three latent components (what the analysis model actually sees as the outcome) is, under the separable form, exactly a random intercept plus an AR(1) residual process — precisely `corCAR1()`. Under the non-separable published structure, the summed outcome instead carries an extra white-noise term the analysis model does not represent (`docs/35`, Section 4.6). Generating data under one correlation model while analyzing it as though a different one held is an avoidable mismatch; adopting the separable form removes it.

Faster computation. Inverting and Cholesky-factorizing a Kronecker-product matrix reduces to factorizing the $p \times p$ and $n \times n$ pieces separately, rather than the full $np \times np$ matrix.

6. What this costs

Old results do not reproduce exactly. Because the lagged cross-component correlations change value (not just the within- component ones), no simulation result computed under the published structure reproduces exactly under the separable one; this is intentional, not an error (`docs/35`, Section 5, item 1).

One decay rate for every component. The separable form uses a single ρ for TV, PB and BR alike. It cannot represent, for instance, a placebo-belief effect that fades on its own, faster or slower timescale than the drug-response component does. A valid extension (a sum of several separable terms, one per rate) exists but gives up the closed-form ceiling (`docs/35`, Section 8).

No same-visit excess correlation. Under the separable form, the ratio of cross-component to within-component correlation is the fixed constant c_1 at *every* lag, including zero; a same-visit "shared shock" stronger than this ratio implies cannot be represented without an explicit additional term.

Separability is itself a testable, not a proven, assumption. Whether the three latent components actually satisfy it cannot be checked directly on real data, since only their sum (the observed symptom score) is measured; the components themselves are latent. Adopting separable AR(1) is a stated modeling commitment, consistent with standard multivariate repeated-measures practice, not an empirically validated fact about this particular condition (`docs/35`, Section 5, item 5).

7. Where this leaves the simulations

Separable AR(1) (Kronecker-product correlation, one shared time-decay factor across all three components) replaces the published process's non-separable compound-symmetry structure, as part of the refined process paper 01 recommends. This is one of several bundled changes; `docs/53` covers the carryover-decay fix and `docs/55` the biomarker-coupling fix, both independent of the covariance-structure choice discussed here.

References

- `analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`, Methods and "Room for larger effect sizes" sections, for ρ_{cs} as published (0.8) and as implemented (0.7), and the ceiling table.
- `docs/35-separable-response-covariance.md`, the full technical derivation: the Kronecker/separability argument (Section 4), the advantages-and-disadvantages discussion (Section 5), the ceiling calculation (Section 6), and the per-component decay-rate alternative (Section 8).
- `docs/series/ch10-refined-process.md`, for the refined process's adoption of separable AR(1) and the quoted ceiling figures.
- `docs/53-carryover-recursion-in-published-code.md` and `docs/55-step-coupling-in-published-code.md`, companion notes on the other two defects in the published code.
- `docs/61-cross-factor-correlation-bug-paper02-driver.md`, on a later implementation bug in one driver's cross-component correlation parameters, unrelated to the structural choice discussed here but concerning the same parameters.